

Kola Peninsula Tundra

A treeless landscape of harsh beauty with permafrost, clear lakes, and rivers, and a vital home for Arctic plant and animal species



Northern Russia’s Kola Peninsula is located almost completely within the Arctic Circle, between the Barents and White Seas. Tundra covers nearly all of its 40,000 square miles (100,000 square km), but approximately 22,700 square miles (58,800 square km) are classified as critically endangered coastal Arctic tundra. The landscape is covered with moss, lichen, grasses, and wildflowers, with shrubs such as dwarf Arctic birch and cloudberry; the frozen subsoil known as permafrost (see panel, right) prevents trees from growing. Kola winters are long and extreme; frost often occurs from August through to the following June, and strong winds blow up to 120 days a year. Despite its harsh environment, the Kola’s tundra, lakes, and river systems are home to 200 species of birds and 32 mammal species, including migrating herds of reindeer. However, Soviet nuclear testing and mining have already harmed what was once a pristine environment.

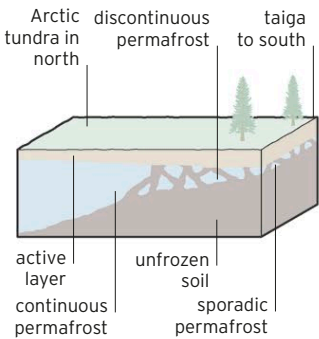


▷ **FRUITFUL SURVIVOR**
Cloudberry can survive temperatures as low as -40°F (-40°C). Their sweet fruit provides food for birds and mammals in fall.



PERMAFROST

Permafrost is ground that has been frozen for two years or more. It usually forms above a layer of unfrozen soil and beneath a layer, called the active layer, that freezes and thaws each year. The permafrost itself may be continuous or discontinuous, or form sporadic pockets.



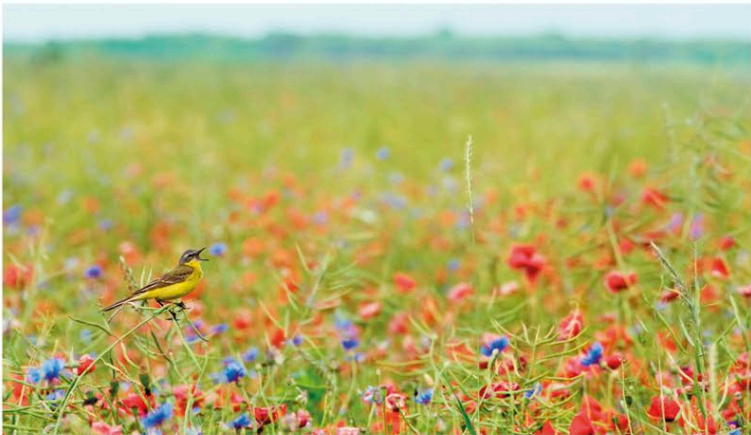
◁ **LIFE IN SMALL DOSES**
Tundra plants remain short to avoid freezing winds. Their shallow roots cannot penetrate permafrost.

The Western Steppe

A flat, windswept plain stretching thousands of miles and forming part of the largest temperate grassland on the planet



The Western Steppe lies chiefly in Ukraine, Russia, and Kazakhstan. With the Altai Mountains as its eastern boundary, it forms half of the vast Eurasian Steppe, the largest temperate grassland on Earth. Steppes get enough precipitation for grass growth but not enough for trees. The Western Steppe is 2,500 miles (4,000 km) long and is often called a “sea of grass,” although where rivers and streams cut through the otherwise unbroken landscape, some trees do grow. Bordered to the north by taiga, and to the south by desert, the Western Steppe is relatively mild compared to eastern grasslands, but is still subjected to howling winds and temperatures ranging from 80°F (27°C) during the day to below freezing at night.



◁ **GRASSLAND FOOD CHAIN**
The grasses and wildflowers of the region abound with insects such as grasshoppers and beetles, which in turn attract many birds, including the western yellow wagtail.